CSMMetal Bio-Sync Seat Cushion

Cost Analysis per Unit — CSMFAB MAG001

Category: Aegis Embark — Seating **Unit Basis:** 1 standard seat cushion **Date:** June 2026 | **Document Type:** CSMFAB Cost Study V2.0

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1. Product Summary

This cost analysis is derived from the V2.0 specification and v1to2 versioning document for this product. All material costs reflect 2026 market pricing. Three production scales are costed: - **Prototype:** 1-5 units (single-setup, high labor, no tooling amortization) - **Small Batch:** 100-500 units/year (semi-tooled, moderate labor) - **Production Volume:** 5,000-50,000 units/year (fully tooled, optimized labor)

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2. Bill of Materials (BOM)

Component	Quantity/Unit	Unit Price	Extended Cost
LORD MRF-140CG adaptive bladder	0.3 kg	$250.00/kg	$75.00
PVDF-TrFE harvesting layer (12 elements)	0.03 kg	$350.00/kg	$10.50
Polyimide-silica aerogel thermal layer (8mm)	0.3 kg	$250.00/kg	$75.00
PVDF bio-feedback film sensors (heart rate/respir.)	0.01 kg	$350.00/kg	$3.50
KNbO3-BaTiO3 Schumann resonator base	0.01 kg	$180.00/kg	$1.80
YInMn Blue Aegis-C outer shell	0.3 kg	$75.00/kg	$22.50
SmCo electromagnet (MRF control)	0.05 kg	$120.00/kg	$6.00
BFRP shell (Elium®)	1 kg	$25.00/kg	$25.00
BOM Subtotal			**$219.30**

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3. Labor, Process & Overhead Costs

Cost Element	Prototype	Small Batch	Production Volume
BOM Materials	$219.30	$179.83	$142.55
Direct Labor	$210.00	$140.00	$90.00
Overhead (15% of labor)	$31.50	$21.00	$13.50
Quality Control & Testing	$25.20	$11.20	$4.50
Packaging & Logistics	$25.00	$20.00	$15.00

Tooling & Equipment Notes: MRF bladder mold: $6,000 NRE. Harvester stack: $28/unit.

4. Per-Unit Cost Summary

Scale	Total COGS	Suggested MSRP Range	Gross Margin
Prototype (1-5 units)	**$920**	$1,472 – $2,024	35-55%
Small Batch (100-500/yr)	**$620**	$930 – $1,240	33-50%
Production Volume (5K-50K/yr)	**$440**	$616 – $792	28-45%

Market Reference: $520-760 (seat cushion wellness market $50-400; significant health monitoring premium)

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5. Key Cost Drivers (V2.0)

Driver	Impact	Mitigation
ZrB2/Si3N4 UHTC ceramics	HIGH — specialty powder $350+/kg	Bulk procurement; domestic synthesis pilot (Japan NIMS CO2 route)
MXene Ti3C2Tx	VERY HIGH — $2,000/kg at 2026 scale	Tile pattern minimizes usage; price projected -60% by 2028 as production scales
Elium® thermoplastic BFRP	MEDIUM — recyclability adds value; $12/kg resin	Phoenix Protocol recycling recovers 100% monomer; net cost offset
Tooling NRE	HIGH for first production run	Paid by pre-orders or grant funding; amortized quickly
Flash Sintering process	MEDIUM — shares SPS equipment	Regional ceramic sintering facility sharing model

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6. Phoenix Protocol Circular Economy Value

At end-of-life, material recovery adds back-value: - **Recovered material value per unit:** $39 – $76 - **Recovery reduces net lifecycle cost by:** 15-30% - **Critical minerals (In, Y, Co, Ti) recovered** via ionic liquid extraction

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End of CSMMetal Bio-Sync Seat Cushion Cost Analysis | CSMFAB MAG001 | June 2026